

ASSIGNMENT NO.3: Conditional Queries and Data Filtering

Q1) Create a database named JuniorCollegeDB and a table named Student. The table should contain the following columns: Column Name Data Type Constraint StudentID INT PRIMARY KEY StudentName VARCHAR(50) Age INT Course VARCHAR(30) City VARCHAR(30) Marks INT Email VARCHAR(60) Insert at least 10 records into the table. Some records should contain NULL values in the Email column.

ANS:

```
CREATE DATABASE JuniorCollegeDB;
```

```
USE JuniorCollegeDB;
```

```
CREATE TABLE Student (
```

```
    StudentID INT PRIMARY KEY,
```

```
    StudentName VARCHAR(50),
```

```
    Age INT,
```

```
    Course VARCHAR(30),
```

```
    City VARCHAR(30),
```

```
    Marks INT,
```

```
    Email VARCHAR(60)
```

```
);
```

```
INSERT INTO Student (StudentID, StudentName, Age, Course, City, Marks, Email)
```

```
VALUES
```

```
(1, 'Aarav Shah', 18, 'BSc IT', 'Mumbai', 85, 'aarav@email.com'),
```

```
(2, 'Meera Joshi', 19, 'BCA', 'Pune', 75, NULL),
```

```
(3, 'Rohan Patil', 20, 'BSc IT', 'Nashik', 92, 'rohan@email.com'),
```

```
(4, 'Ananya Desai', 19, 'BSc', 'Mumbai', 81, 'ananya@email.com'),
```

```
(5, 'Karan Mehta', 21, 'BCA', 'Pune', 68, NULL),
```

```
(6, 'Neha Sharma', 18, 'BSc IT', 'Mumbai', 88, 'neha@email.com'),
```

```
(7, 'Aditya Verma', 20, 'BSc', 'Nashik', 72, NULL),
```

(8, 'Priya Singh', 19, 'BCA', 'Pune', 95, 'priya@email.com'),
 (9, 'Rahul Rao', 21, 'BSc IT', 'Mumbai', 79, 'rahul@email.com'),
 (10, 'Sneha Iyer', 20, 'BSc', 'Nashik', 84, NULL);

The first screenshot shows the phpMyAdmin interface with the following SQL queries and results:

```

CREATE DATABASE JuniorCollegeDB;

USE JuniorCollegeDB;

CREATE TABLE Student ( StudentID INT PRIMARY KEY, StudentName VARCHAR(50), Age INT, Course VARCHAR(30), City VARCHAR(30), Marks INT, Email VARCHAR(60) );

INSERT INTO Student (StudentID, StudentName, Age, Course, City, Marks, Email) VALUES (1, 'Aarav Shah', 18, 'BSc IT', 'Mumbai', 85, 'aarav@email.com'), (2, 'Meera Joshi', 19, 'BCA', 'Pune', 75, NULL), (3, 'Rohan Patil', 20, 'BSc IT', 'Nashik', 92, 'rohan@email.com'), (4, 'Ananya Desai', 19, 'BSc', 'Mumbai', 81, 'ananya@email.com'), (5, 'Karan Mehta', 21, 'BCA', 'Pune', 68, NULL), (6, 'Neha Sharma', 18, 'BSc IT', 'Mumbai', 88, 'neha@email.com'), (7, 'Aditya Verma', 20, 'BSc', 'Nashik', 72, NULL), (8, 'Priya Singh', 19, 'BCA', 'Pune', 95, 'priya@email.com'), (9, 'Rahul Rao', 21, 'BSc IT', 'Mumbai', 79, 'rahul@email.com'), (10, 'Sneha Iyer', 20, 'BSc', 'Nashik', 84, NULL);
  
```

The second screenshot shows the resulting table with 10 rows:

StudentID	StudentName	Age	Course	City	Marks	Email
1	Aarav Shah	18	BSc IT	Mumbai	85	aarav@email.com
2	Meera Joshi	19	BCA	Pune	75	NULL
3	Rohan Patil	20	BSc IT	Nashik	92	rohan@email.com
4	Ananya Desai	19	BSc	Mumbai	81	ananya@email.com
5	Karan Mehta	21	BCA	Pune	68	NULL
6	Neha Sharma	18	BSc IT	Mumbai	88	neha@email.com
7	Aditya Verma	20	BSc	Nashik	72	NULL
8	Priya Singh	19	BCA	Pune	95	priya@email.com
9	Rahul Rao	21	BSc IT	Mumbai	79	rahul@email.com
10	Sneha Iyer	20	BSc	Nashik	84	NULL

Q2) Write a MySQL query to display students who are from Mumbai and have marks greater than 80. The output should include: Column StudentID StudentName City Marks

ANS:

```

SELECT StudentID,
       StudentName,
       City,
       Marks
FROM student
  
```

WHERE City = 'Mumbai'

AND Marks > 80;

The screenshot shows the phpMyAdmin interface. The left sidebar displays a database structure with 'juniorcollegedb' selected. The main panel shows a query result for the 'student' table. The query is: `SELECT StudentID, StudentName, City, Marks FROM student WHERE City = 'Mumbai' AND Marks > 80;`. The result shows 3 rows:

StudentID	StudentName	City	Marks
1	Aarav Shah	Mumbai	85
4	Ananya Desai	Mumbai	81
6	Neha Sharma	Mumbai	88

Q3) Write a MySQL query to display students who belong to either Pune or Nashik. Use the OR operator and display the following columns:

Column StudentName Course City

ANS:

SELECT StudentName, Course, City

FROM student

WHERE City = 'Pune'

OR City = 'Nashik';

The screenshot shows the phpMyAdmin interface. The left sidebar displays a database structure with 'juniorcollegedb' selected. The main panel shows a query result for the 'student' table. The query is: `SELECT StudentName, Course, City FROM student WHERE City = 'Pune' OR City = 'Nashik';`. The result shows 6 rows:

StudentName	Course	City
Meera Joshi	BCA	Pune
Rohan Patil	BSc IT	Nashik
Karan Mehta	BCA	Pune
Aditya Verma	BSc	Nashik
Priya Singh	BCA	Pune
Sneha Iyer	BSc	Nashik

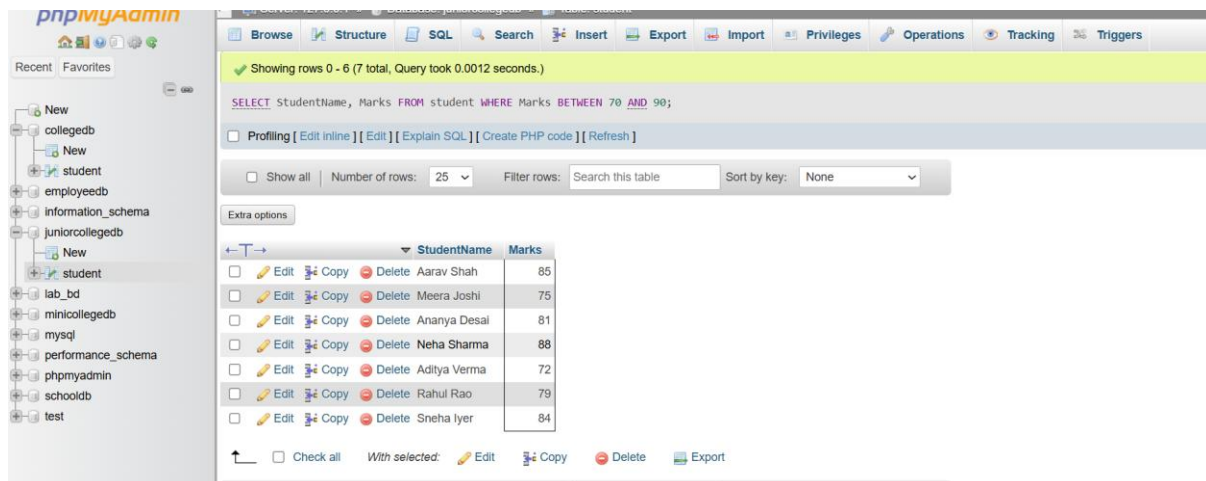
Q4) Write a MySQL query to display students whose marks are between 70 and 90. Use the BETWEEN operator and display: Column StudentName
Marks

ANS:

SELECT StudentName, Marks

FROM student

WHERE Marks BETWEEN 70 AND 90;



Showing rows 0 - 6 (7 total, Query took 0.0012 seconds.)

```
SELECT StudentName, Marks FROM student WHERE Marks BETWEEN 70 AND 90;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

	StudentName	Marks
☐	Aarav Shah	85
☐	Meera Joshi	75
☐	Ananya Desai	81
☐	Neha Sharma	88
☐	Aditya Verma	72
☐	Rahul Rao	79
☐	Sneha Iyer	84

Check all | With selected: Edit Copy Delete Export

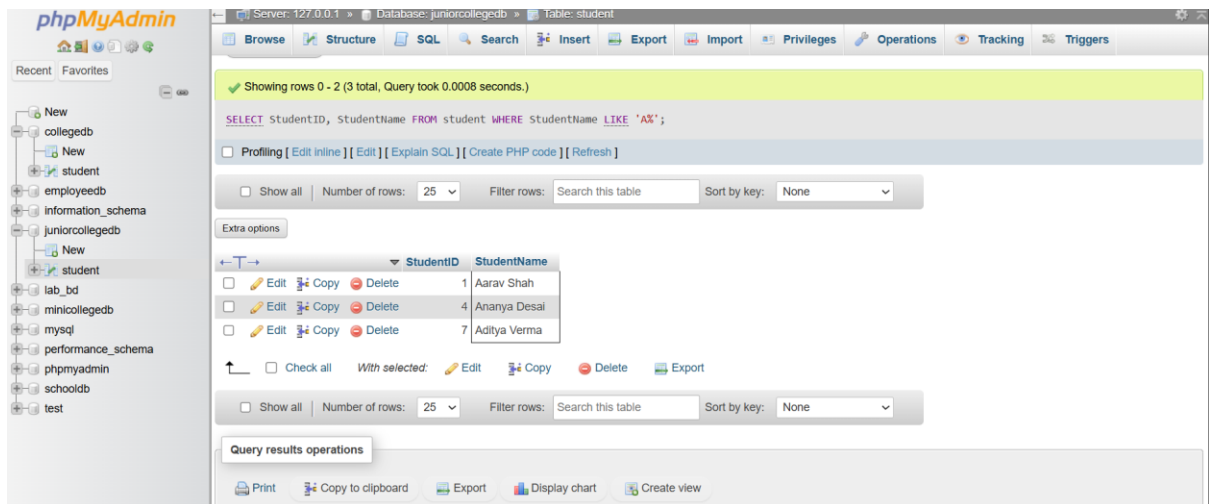
Q5) Write a MySQL query to display students whose names start with the letter A. Use the LIKE operator and display: Column StudentID
StudentName

ANS:

SELECT StudentID, StudentName

FROM student

WHERE StudentName LIKE 'A%';



Q6) Write a MySQL query to display students whose email address is missing. Use the IS NULL operator and display

COLUMN
STUDENTID
STUDENTNAME
EMAIL

ANS:

SELECT StudentID, StudentName, Email

FROM student

WHERE Email IS NULL;

